

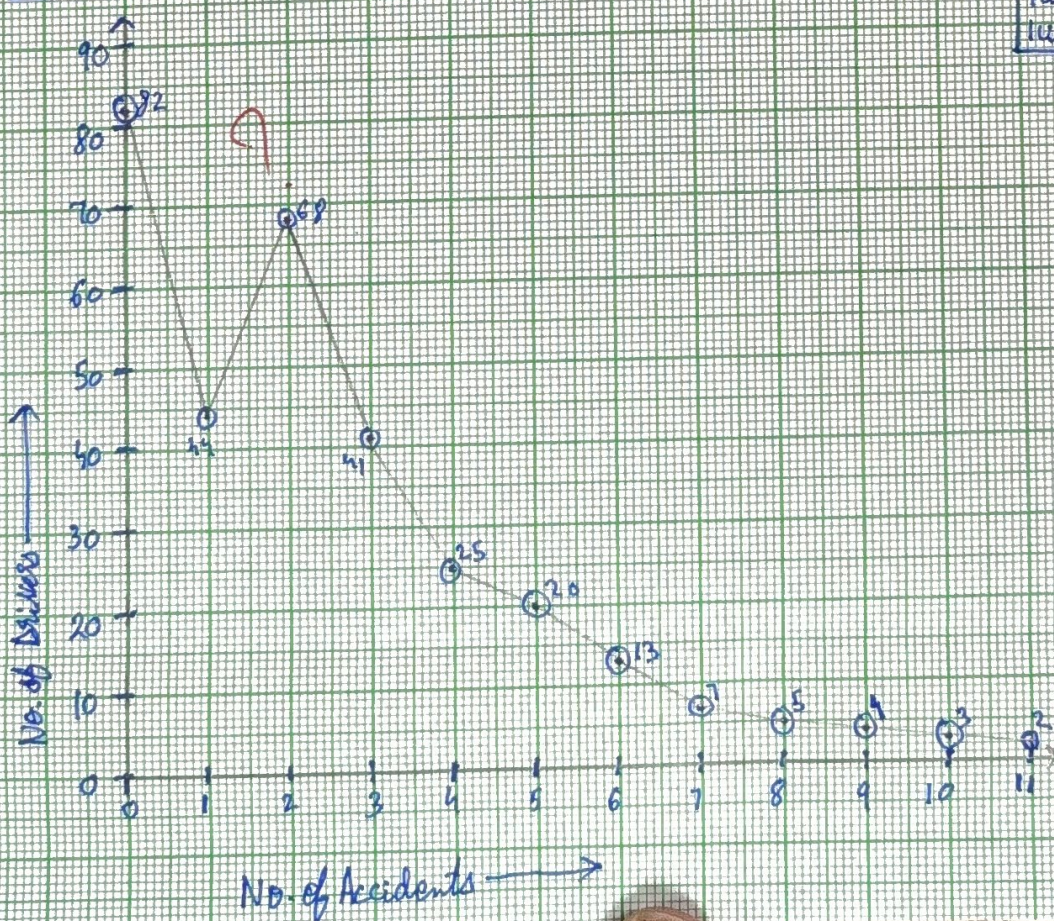
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Managing Uncertainty Assignment-1

Q1 The following data shows the number of accidents sustained by 314 drivers of a public utility company over a period of 5 years.
Represent the data by a line diagram.

No. of Accidents	No. of Drivers
0	82
1	44
2	68
3	41
4	25
5	20
6	13
7	7
8	5
9	4
10	3
11	2

Ans 1

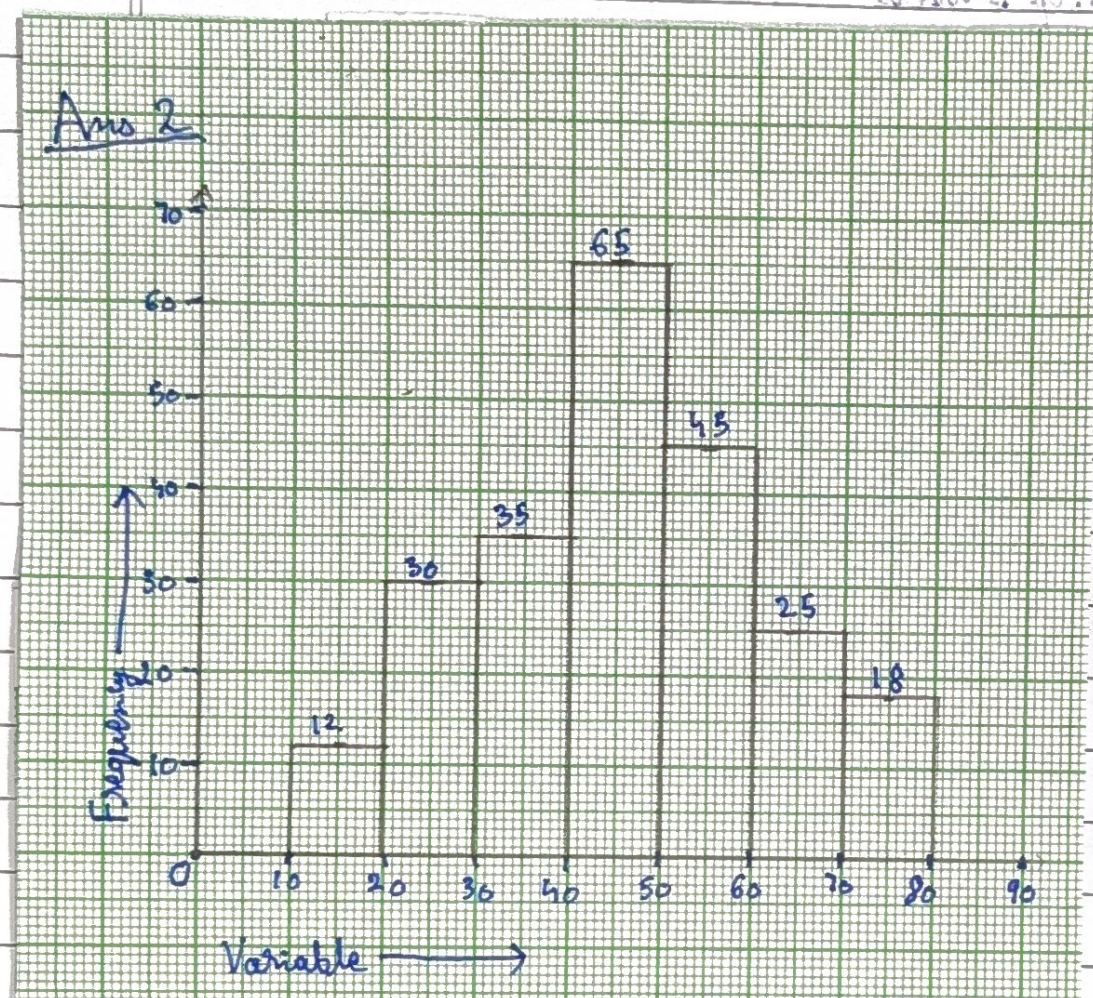


Scale:
1 unit on X-axis = 1 Accident
1 unit on Y-axis = 10 Drivers

Q2

Draw histogram for the following distribution.

Variable:	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Frequency:	12	30	35	65	45	25	18



83

Draw a pie-chart for the following data

Items	Agriculture & Rural Development	Industries & Urban Development	Health Education	Miscellaneous
Proposed Expenditure (in million Rs.)	4200	1500	1000	500

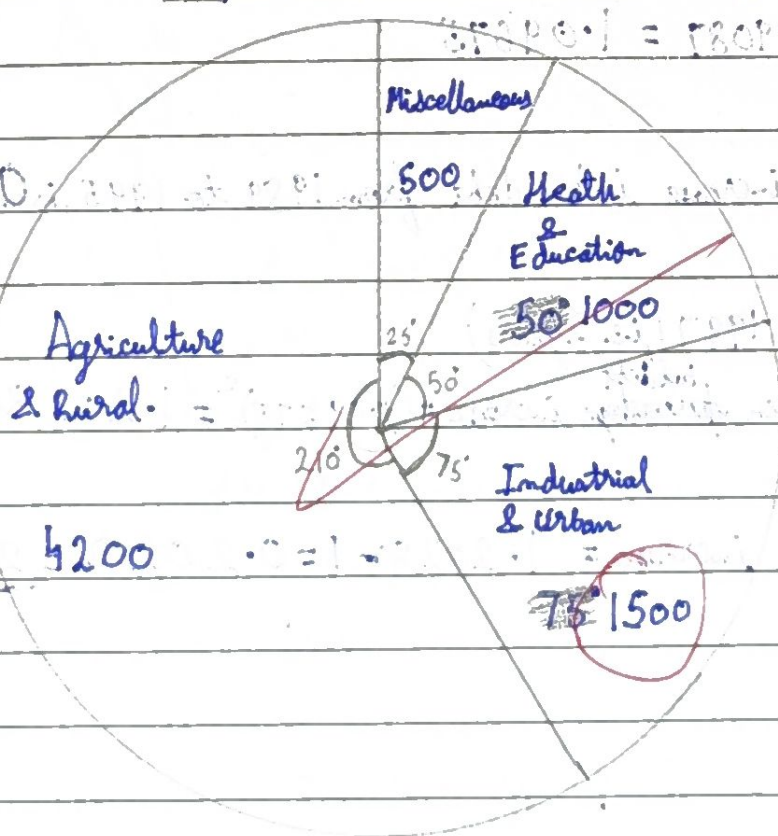
$$\text{Total Expenditure} = 4200 + 1500 + 1000 + 500 = 7200$$

$$\text{Angle} = \frac{\text{Item Expenditure} \times 360^\circ}{\text{Total Expenditure}}$$

$$\text{Agri. \& Rural} = \frac{4200 \times 360^\circ}{7200} = 210^\circ ; \quad \text{Health \& Edu.} = \frac{1000 \times 360^\circ}{7200} = 50^\circ$$

$$\text{Industrial \& Urban} = \frac{1500 \times 360^\circ}{7200} = 75^\circ ; \quad \text{Miscellaneous} = \frac{500 \times 360^\circ}{7200} = 25^\circ$$

Proposed Expenditure (in million Rs.)



Q4

Branch	Current sales (w)	Projected Growth (x)	w × x
Atlantic branch	\$193.8 million	7.25 %	1405.05
Midwest branch	\$79.3 million	8.2 %	650.26
Pacific branch	\$57.5 million	7.15 %	411.125
$\Sigma w = 330.6$			$\Sigma (w \times x) = 2466.435$

$$\text{Avg. rate of Growth} = \text{Weighted mean of data} = \frac{\Sigma (w \times x)}{\Sigma w} = \frac{2466.435}{330.6} = 7.46$$

$\therefore$ Avg. forecasted rate of sales growth = 7.46%

Q5

1989 1990 1991 1992 1993 1994 1995

0.11, 0.09, 0.075, 0.08, 0.095, 0.108, 0.120

Growth factor (1+r) for each year: 1.11, 1.09, 1.075, 1.08, 1.095, 1.108, 1.120

$$\text{Geometric mean} = \sqrt[7]{1.11 \times 1.09 \times 1.075 \times 1.08 \times 1.095 \times 1.108 \times 1.120}$$

$$= \sqrt[7]{1.9087} = 1.09675$$

hence, avg. percentage increase in bad debt from 1989 to 1995 is $0.09675 = 9.675\%$

If this continues till 1997 (i.e. 2 years)

$$\text{Then } \text{bad debt percentage increase} = (1.09675)^2 = 1.20286$$

$$\text{percentage increase} = 1.20286 - 1 = 0.20286 = \underline{20.29\%}$$

Q 6

Cost of Product over 4 months: \$300, \$275, \$250, \$240 and \$225

To calculate: Average Rate of Decrease.

$$\text{Ratio of final cost to initial cost} = \frac{275}{300} \times \frac{250}{275} \times \frac{240}{250} \times \frac{225}{240} = \frac{225}{300} = 0.75$$

$$\text{Average monthly Ratio} = G.M. = \sqrt[4]{0.75} = 0.9306$$

$$\text{Rate of Decrease} = 1 - 0.9306 = 0.06939 = 6.94\% \text{ per month}$$

Q 7

a) Frequency distribution:

Interval	Frequency
15-19	10 ← Modal Class
20-24	9
25-29	3
30-34	4
35 <	9

$$b) \text{ Modal Value: } M_0 = L_{m0} + \left(\frac{d_1}{d_1 + d_2} \right) w$$

Where, L_{m0} = Lower limit of Modal class = 15

$$d_1 = \text{frequency of Modal class} - \text{frequency of class above} = 10 - 0 = 10$$

$$d_2 = \text{freq. of Modal class} - \text{freq. of class below} = 10 - 9 = 1$$

$$w = \text{Class interval width} = 19 - 15 = 4$$

$$\Rightarrow M_0 = 15 + \left(\frac{10}{11} \right) 4 = 15 + 3.636 = 19.55 = \text{Mode}$$

C

Find mean of raw data

$$\bar{x} = \frac{19+17+15+20+23+41+33+21+19+20+18+33+32+29+29+19+18+20+17+22+55+11+22+25+28+30+44+19+20+39}{30}$$

$$= \frac{760}{30} = 25.33$$

d

Compare the mean and mode; Comment on which of the two is a better central tendency of the data and why.

Comparing the mean ^(25.33) and mode ^(19.55), we see that mean is greater than mode.

Mode is a better representation as it is not influenced by the extreme values or outliers in the data, and provides a more accurate representation of ages of students.

Q8

Age (years)	No. of persons (in thousands) (C.F)	(Frequency)
Below 10	2	2
10 - 20	5	3
20 - 30	9	4
30 - 40	12	3
40 - 50	14	2
50 - 60	15	1
60 - 70	15.5	0.5
70 and above	15.6	0.1

$$N = 15.6$$

$$\frac{N}{2} = \frac{15.6}{2} = 7.8$$

(i) Find Median

$$\text{Median} = \left(\frac{\frac{N}{2} - F}{f_m} \right) w + L_m$$

Where, N = Total no. of items in distribution = 15.6

F = Cumulative frequency of the class above before median class = 5.8

f_m = Frequency of median class = 4

w = class interval width = 10

L_m = Lower limit of median class = 20

$$\tilde{m} = \frac{7.8 - 5.8}{4} \times 10 + 20 = \frac{2.0}{4} \times 10 + 20 = 25$$

(ii) Why is median more suitable

The median is more suitable central tendency as here the final age group is an open class and it is impossible to find the midpoint; hence the mean cannot be accurately found.

Also

Q9

Find missing frequency, given the median is Rs 2400

Sales (x)	No. of shops (f)	C.F.
0-1000	5	5
1000-2000	25	30
2000-3000	x	30+x ← Median Class
3000-4000	18	48+x
4000-5000	7	55+x

$$\text{Median} = \left(\frac{\frac{N}{2} - F}{f_m} \right) w + L_m$$

$$\text{Median} = 2400$$

$$N = 55 + x$$

$$F = 30$$

$$f_m = x$$

$$w = 1000$$

$$L_m = 2000$$

$$2400 = \left(\frac{\frac{55+x}{2} - 30}{x} \right) 1000 + 2000$$

$$400 = \frac{(x-5)1000}{2x}$$

$$800x = 1000x - 5000$$

$$200x = 5000$$

$$x = 25$$

hence, missing frequency = 25.

Q10

Find Missing Frequencies, given the median is 50, and total observations are 100.

Expenditure	No. of families	C.F.
0-20	14	14
20-40	f_1	$14+f_1$
40-60	27 → Median Class	$41+f_1$
60-80	f_2	$41+f_1+f_2$
80-100	15	$56+f_1+f_2$
$\Sigma f = 100$		

$$\text{Median} = \left(\frac{\frac{N}{2} - F}{f_m} \right) w + L_m ; \text{ where: Median} = 50$$

$$N = 100$$

$$F = 14 + f_1$$

$$f_m = 27$$

$$w = 20$$

$$L_m = 40$$

$$50 = \left(\frac{50 - 14 - f_1}{27} \right) 20 + 40$$

$$\frac{10 \times 27}{20} = 36 - f_1$$

$$13.5 = 36 - f_1$$

$$f_1 = 22.5$$

$$N = 56 + f_1 + f_2 = 100$$

$$f_2 = 100 - 56 - 22.5$$

$$f_2 = 21.5$$

As the frequency represent families, they need to be whole numbers. hence rounding off and adjusting values we get, $f_1 = 23$ and $f_2 = 21$.